

The linear canonical wavelet transform on some function spaces

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It is well known that the domain of Fourier transform (FT) can be extended to the Schwartz space $\mathcal{S}(R)$ for convenience. As a generation of FT, it is necessary to detect the linear canonical transform (LCT) on a new space for obtaining the similar properties like FT on $\mathcal{S}(R)$. Therefore, a space $\mathcal{S}_{A_1}(R)$ generalized from $\mathcal{S}(R)$ is introduced firstly, and further we prove that LCT is a homeomorphism from $\mathcal{S}_{A_1}(R)$ onto itself. The linear canonical wavelet transform (LCWT) is a newly proposed transform based on the convolution theorem in LCT domain. Moreover, we propose an equivalent definition of LCWT associated with LCT and further study some properties of LCWT on $\mathcal{S}_{A_1}(R)$. Based on these properties, we finally prove that LCWT is a linear continuous operator on the spaces of L^{p,A_1} and $H_{A_1}^{s,p}$.

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1. Introduction

Fourier transform (FT) is frequently introduced as an transform on $L^1(R)$, as well as inversion Fourier transform. Under this assumption, the Fourier transform of many functions are not convergence or have no meanings, such as the general functions on $L^p(R)$ ($p > 1$) and $f(x) = 1$. The generalized functions or distributions are considered to enlarge the domain of FT. The strategy is that consider $C_0^\infty(R) \subset L^p(R)$ as the domain of FT. However, FT does not map $C_0^\infty(R)$ to $C_0^\infty(R)$ because the FT of an element in $C_0^\infty(R)$ cannot vanish on an open set. Therefore, the

domain of FT is extended to the space of Schwartz functions that slightly larger than $C_0^\infty(R)$. The rapid decay of Schwartz functions at infinity allows us to develop its fundamental properties without encountering any convergence problems. FT on Schwartz space $\mathcal{S}(R)$ is defined as⁶:

$$F_\phi(u) = \mathcal{F}[\phi(t)](u) = \frac{1}{\sqrt{2\pi}} \int_R \phi(t) e^{-iut} dt, \quad \forall \phi(t) \in \mathcal{S}(R), \quad (1.1)$$

where the Schwartz space $\mathcal{S}(R) = \{\phi \in C^\infty | \sup_{t \in R} |t^\alpha D^\beta \phi(t)| < \infty, \forall \alpha, \beta \in N\}$. It is a Frechet space with the seminorms defined as $\|\phi\|_{\mathcal{S}(R)} = \sup_{t \in R} |t^\alpha D^\beta \phi(t)|$.

Some properties of FT on Schwartz space $\mathcal{S}(R)$ can be obtained and shown in following lemma.

Lemma 1.1. *Let $\phi(t), \varphi(t) \in \mathcal{S}(R), \beta \in N$, D and $*$ are ordinary differential operator and convolution operator respectively, then*

- (i) $\mathcal{F}[D^\beta \phi(t)](u) = (iu)^\beta \mathcal{F}[\phi(t)](u)$.
- (ii) $\mathcal{F}[t^\beta \phi(t)](u) = i^\beta D_u^\beta \mathcal{F}[\phi(t)](u)$.
- (iii) $\mathcal{F}[\phi(t) * \varphi(t)](u) = \mathcal{F}[\phi(t)](u) \cdot \mathcal{F}[\varphi(t)](u)$.
- (iv) $\int_R \phi(t) \overline{\varphi(t)} dt = \int_R \mathcal{F}[\phi(t)](u) \overline{\mathcal{F}[\varphi(t)](u)} du$.
- (v) $\int_R |\phi(t)|^2 dt = \int_R |\mathcal{F}[\phi(t)](u)|^2 du$.

Based on these properties, a well known theorem about FT on $\mathcal{S}(R)$ can be shown as follows.⁶

Lemma 1.2. *The FT is a homeomorphism from $\mathcal{S}(R)$ onto itself.*

Recently, many researchers focus on the FT and generalized FT on other spaces.^{2,7,8} The linear canonical transform (LCT) is a generalization of FT, and it can also be reduced to the fractional Fourier transform (FRFT), fresnel transform (FST) and scale transform by choosing suitable parameters. At present, some well known concepts, such as the uncertainty principles, the convolution and product theorems, and the Hilbert transform have been detected in the LCT domain.^{1,23} LCT has been shown to be one of the important tools in the optics, quantum mechanics and signal processing fields.³⁻⁵ In follows, some properties of LCT on the space $\mathcal{S}_{A_1}(R)$ are detected.

2. Linear Canonical Transform

2.1. Definition

Definition 2.1. The LCT of $\phi(t)$ with real parameters $A_1 = (a_1, b_1, c_1, d_1)$ is defined as

$$L_\phi^{A_1}(u) = \mathcal{L}^{A_1}[\phi(t)](u) = \int_R \phi(t) K_{A_1}(t, u) dt \quad (2.1)$$

Table 1. Some special cases of LCT.

Parameter	Representation
$A_1 = (0, 1, -1, 0)$	FT: $L_\phi^{A_1}(u) = \frac{1}{\sqrt{i2\pi}} \int_R \phi(t) e^{-iut} dt$
$A_1 = (1, \lambda z/(2\pi), 0, 1)$	FST: $L_\phi^{A_1}(u) = \frac{1}{\sqrt{i2\pi\lambda}} \int_R \phi(t) e^{\frac{i\pi}{\lambda z}(u-t)^2} dt$
$A_1 = (\cos \theta, \sin \theta, -\sin \theta, \cos \theta)$	FRFT: $L_\phi^{A_1}(u) = \frac{1}{\sqrt{i2\pi \sin \theta}} \int_R \phi(t) e^{\frac{i}{2}[(u^2+t^2) \cot \theta - 2ut \csc \theta]} dt$

with the kernel

$$K_{A_1}(t, u) = \begin{cases} \frac{1}{\sqrt{i2\pi b_1}} \exp\left(i \frac{a_1 t^2 + d_1 u^2 - 2tu}{2b_1}\right) & b_1 \neq 0, \\ \sqrt{d_1} \exp\left(i \frac{c_1 d_1 u^2}{2}\right) \delta(t - d_1 u) & b_1 = 0, \end{cases} \quad (2.2)$$

where $A_1 \in S(2, R) = \{A \in M(2, R) \mid \det(A) = 1\}$, $M(2, R)$ is a set of 2×2 real matrix. $L_\phi^{A_1}(u)$ denotes the LCT of $\phi(t)$ and $\mathcal{L}^{A_1}$ denotes the LCT operator. When $b_1 \neq 0$, the inverse LCT is given by

$$\phi(t) = \mathcal{L}^{A_1^{-1}}[L_\phi^{A_1}(u)](t) = \int_R L_\phi^{A_1}(u) K_{A_1}^*(t, u) du, \quad (2.3)$$

where the kernel $K_{A_1}^*(t, u) = K_{A_1^{-1}}(t, u)$, $A_1^{-1} = (d_1, -b_1, -c_1, a_1)$.

Since LCT is a three parameters class of linear integral transform, LCT can be seen as a generations of FT, FRFT and FST with some appropriate parameters. The details are listed on the Table 1.

2.2. The LCT on $\mathcal{S}_{A_1}(R)$

Definition 2.2. The test function space $\mathcal{S}_{A_1}(R)$ is defined as

$$\mathcal{S}_{A_1}(R) = \left\{ \phi \in C^\infty \left| \sup_{t \in R} |t^\alpha \Delta_t^\beta \phi(t)| < \infty, \forall \alpha, \beta \in N \right. \right\}, \quad (2.4)$$

where $\Delta_t = \frac{d}{dt} - it \frac{a_1}{b_1}$. It is also a Frechet space with the seminorms defined as $\|\phi\|_{\mathcal{S}_{A_1}(R)} = \sup_{t \in R} |t^\alpha \Delta_t^\beta \phi(t)|$. When $a_1 = 0$, $\mathcal{S}_{A_1}(R)$ reduces to Schwartz space $\mathcal{S}(R)$.

Inspired by the corresponding properties of FT, some properties of LCT on $\mathcal{S}_{A_1}(R)$ are studied in Theorem 2.1.

Theorem 2.1. Assume $\forall \phi(t) \in \mathcal{S}_{A_1}(R), \beta \in N, \Delta_t^* = -(\frac{d}{dt} + it \frac{a_1}{b_1})$, then

- (i) $\Delta_t^\beta K_{A_1}(t, u) = (-iu/b_1)^\beta K_{A_1}(t, u)$.
- (ii) $\int_R \Delta_t^\beta K_{A_1}(t, u) \phi(t) dt = \int_R K_{A_1}(t, u) (\Delta_t^*)^\beta \phi(t) dt$.
- (iii) $\int_R (\Delta_t^*)^\beta K_{A_1}(t, u) \phi(t) dt = \int_R K_{A_1}(t, u) \Delta_t^\beta \phi(t) dt$.

- (iv) $\mathcal{L}^{A_1}[(\Delta_t^*)^\beta \phi(t)](u) = (-iu/b_1)^\beta \mathcal{L}^{A_1}[\phi(t)](u).$
 (v) If $a_1 = d_1$, then $\mathcal{L}^{A_1}[t^\beta \phi(t)](u) = (ib_1)^\beta \Delta_u^\beta \mathcal{L}^{A_1}[\phi(t)](u).$
 (vi) $\mathcal{L}^{A_1}[\phi(t)\Theta_{A_1}\varphi(t)](u) = \sqrt{2\pi}L_\phi^{A_1}(u)F_\varphi(u/b_1)$, where $\phi(t)\Theta_{A_1}\varphi(t) = \int_{-\infty}^{+\infty} \times$
 $\varphi(\tau)\phi(t-\tau)e^{-\frac{ia_1}{b_1}\tau(t-\tau/2)}d\tau.$
 (vii) $\int_R \phi(t)\overline{\varphi(t)}dt = \int_R L_\phi^{A_1}(u)\overline{L_\varphi^{A_1}(u)}du.$
 (viii) $\int_R |\phi(t)|^2 dt = \int_R |L_\phi^{A_1}(u)|^2 du.$

Proof. (i) According to the definitions of $K_{A_1}(t, u)$ and Δ_t , then

$$\begin{aligned}\Delta_t K_{A_1}(t, u) &= \left(\frac{d}{dt} - it \frac{a_1}{b_1} \right) \left[\frac{1}{\sqrt{i2\pi b_1}} \exp \left(i \frac{a_1 t^2 + d_1 u^2 - 2tu}{2b_1} \right) \right] \\ &= \frac{1}{\sqrt{i2\pi b_1}} \left[\exp \left(i \frac{a_1 t^2 + d_1 u^2 - 2tu}{2b_1} \right) \left(it \frac{a_1}{b_1} - iu \frac{1}{b_1} - it \frac{a_1}{b_1} \right) \right] \\ &= \left(-\frac{iu}{b_1} \right) K_{A_1}(t, u).\end{aligned}$$

Continuing in this way, we obtain

$$\Delta_t^\beta K_{A_1}(t, u) = \left(-\frac{iu}{b_1} \right)^\beta K_{A_1}(t, u) (\forall \beta \in N). \quad (2.5)$$

(ii) According to the subsection integral, then

$$\begin{aligned}\int_R \Delta_t K_{A_1}(t, u) \phi(t) dt &= \int_R \left(\frac{d}{dt} - it \frac{a_1}{b_1} \right) K_{A_1}(t, u) \phi(t) dt \\ &= \int_R K_{A_1}(t, u) \left[-\left(\frac{d}{dt} + it \frac{a_1}{b_1} \right) \right] \phi(t) dt \\ &= \int_R K_{A_1}(t, u) (\Delta_t^*) \phi(t) dt.\end{aligned}$$

Continuing in this way, the final result can be obtained.

(iii) The proof is similar to that of (ii).

(iv) By using the result of (iii), we have

$$\begin{aligned}\mathcal{L}^{A_1}[(\Delta_t^*)^\beta \phi(t)](u) &= \int_R K_{A_1}(t, u) (\Delta_t^*)^\beta \phi(t) dt \\ &= \int_R \Delta_t^\beta K_{A_1}(t, u) \phi(t) dt \\ &= \left(-\frac{iu}{b_1} \right)^\beta \int_R K_{A_1}(t, u) \phi(t) dt \\ &= \left(-\frac{iu}{b_1} \right)^\beta \mathcal{L}^{A_1}[\phi(t)](u).\end{aligned}$$

(v) By using the results of (iii) and (iv), we have

$$\begin{aligned}(ib_1)^\beta \Delta_u^\beta \mathcal{L}^{A_1}[\phi(t)](u) &= (ib_1)^\beta \int_R \Delta_u^\beta K_{A_1}(t, u) \phi(t) dt \\ &= (ib_1)^\beta \int_R \left(-\frac{it}{b_1}\right)^\beta K_{A_1}(t, u) \phi(t) dt \\ &= \int_R K_{A_1}(t, u) [t^\beta \phi(t)] dt \\ &= \mathcal{L}^{A_1}[t^\beta \phi(t)](u).\end{aligned}$$

(vi) The results of (vi)–(viii) can be obtained according to Ref. 20. $\square$

Theorem 2.2. *If parameters $A_1 = (a_1, b_1, c_1, d_1)$ satisfying $a_1 = d_1$, then LCT operator $\mathcal{L}^{A_1} : \phi(t) \rightarrow L_\phi^{A_1}(u)$ is a homeomorphism from $\mathcal{S}_{A_1}(R)$ onto itself.*

Proof. First, we prove $L_\phi^{A_1}(u) \in \mathcal{S}_{A_1}(R)$. Based on the (iv) and (v) properties of Theorem 2.1, we have

$$\begin{aligned}|u^\alpha \Delta_u^\beta L_\phi^{A_1}(u)| &= |u^\alpha \Delta_u^\beta \mathcal{L}^{A_1}[\phi(t)](u)| \\ &= \left| u^\alpha \frac{1}{(ib_1)^\beta} \mathcal{L}^{A_1}[t^\beta \phi(t)](u) \right| \\ &= \left| \left(-\frac{i}{b_1}\right)^{\beta-\alpha} \mathcal{L}^{A_1}[(\Delta_t^*)^\alpha(t^\beta \phi(t))](u) \right| \\ &\leq |b_1|^{|\alpha-\beta|} \int_R |(\Delta_t^*)^\alpha(t^\beta \phi(t))| dt \\ &\leq |b_1|^{|\alpha-\beta|} \sup_{t \in R} |t^2 (\Delta_t^*)^\alpha(t^\beta \phi(t))| < \infty.\end{aligned}$$

It means that $L_\phi^{A_1}(u) \in \mathcal{S}_{A_1}(R)$. From the same inequality, if $\|\phi_n - \phi\|_{\mathcal{S}_{A_1}(R)} = \sup_{t \in R} |t^\alpha \Delta_t^\beta [\phi_n(t) - \phi(t)]| \rightarrow 0$, then $\|L_{\phi_n}^{A_1}(u) - L_\phi^{A_1}(u)\|_{\mathcal{S}_{A_1}(R)} = \sup_{u \in R} |u^\alpha \Delta_u^\beta [L_{\phi_n}^{A_1}(u) - L_\phi^{A_1}(u)]| \rightarrow 0$. Therefore, LCT is a linear continuous mapping from $\mathcal{S}_{A_1}(R)$ onto itself. We also note that the inverse LCT has similar form with LCT, so the inverse LCT is also a linear continuous mapping from $\mathcal{S}_{A_1}(R)$ onto itself. Based on the two results, LCT is a homeomorphism from $\mathcal{S}_{A_1}(R)$ onto itself. $\square$

3. Linear Canonical Wavelet Transform

Fractional wavelet transform (FRWT) is proposed based on FRFT²¹ and further studied in Refs. 12 and 15. It is natural to propose a linear canonical wavelet transform (LCWT) based on the convolution theorem in LCT domain.²² The LCWT not only inherits the advantages of multiresolution analysis of WT, but also has the capability of signal representation in the LCT domain.

3.1. Definition

Definition 3.1. The linear canonical wavelet transform (LCWT) of $f(t)$ with parameters A_1 is defined as²²

$$\begin{aligned} W_f^{A_1}(a, b) &= \mathcal{W}^{A_1}[f(t)](a, b) \\ &= f(t) \Theta_{A_1} \left(\frac{1}{\sqrt{a}} \psi \left(-\frac{t}{a} \right) \right) \\ &= \int_R f(t) \overline{\psi_{a,b,A_1}(t)} dt \\ &= \frac{1}{\sqrt{a}} \int_R f(t) \overline{\psi \left(\frac{t-b}{a} \right)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}} dt, \end{aligned} \quad (3.1)$$

where $\psi_{a,b,A_1}(t) = \frac{1}{\sqrt{a}} \psi \left(\frac{t-b}{a} \right) e^{-\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}}$, $\psi \left(\frac{t-b}{a} \right)$ ($a \in R^+, b \in R$) is the continuous wavelet function. The inverse LCWT is given by

$$f(t) = \frac{1}{2\pi C_\psi} \int_R W_f^{A_1}(a, b) \psi_{a,b,A_1}(t) \frac{1}{a^2} da db, \quad (3.2)$$

where $C_\psi = \int_R \frac{|F_\psi(u)|^2}{u} du < \infty$, $F_\psi(u)$ denotes the FT of $\psi(t)$.

Lemma 3.1. According to the definitions of LCT and LCWT, we have

$$\mathcal{L}^{A_1}[W_f^{A_1}(a, b)] = \mathcal{L}^{A_1} \left[f(t) \Theta_{A_1} \left(\frac{1}{\sqrt{a}} \overline{\psi \left(-\frac{t}{a} \right)} \right) \right] = \sqrt{2\pi a} L_f^{A_1}(u) \overline{F_\psi \left(\frac{au}{b_1} \right)}. \quad (3.3)$$

Therefore, the LCWT can be expressed in terms of the LCT of $f(t)$,²² i.e.

$$W_f^{A_1}(a, b) = \sqrt{2\pi a} \int_R L_f^{A_1}(u) \overline{F_\psi \left(\frac{au}{b_1} \right)} K_{A_1^{-1}}(b, u) du. \quad (3.4)$$

Whenever $a_1=0$, LCWT reduces to the classical WT, and if $a_1 = \cos \theta, b_1 = \sin \theta$, LCWT reduces to FRWT proposed by Shi *et al.*²¹

3.1.1. An equivalent definition of LCWT

Theorem 3.1. An equivalent definition of LCWT associated with LCT is given by Eq. (3.5). In further, this definition is equivalent to the Definition 3.1.

$$\begin{aligned} W_f^{A_1}(a, b) &= \sqrt{a} \int_R L_f^{A_1}(u) \exp \left[-\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) + \frac{ibu}{b_1} + \frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right] \\ &\quad \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au)} du \end{aligned} \quad (3.5)$$

Proof. Since

$$\begin{aligned}
 & \mathcal{L}^{A_1}[\psi_{a,b,A_1}(t)](u) \\
 &= \int_R \frac{1}{\sqrt{i2\pi b_1}} \exp\left(i \frac{a_1 t^2 + d_1 u^2 - 2tu}{2b_1}\right) \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) \\
 &\quad \times \exp\left[-\frac{i}{2}(t^2 - b^2) \cdot \frac{a_1}{b_1}\right] dt \\
 &= \frac{1}{\sqrt{i2\pi b_1}} \exp\left[\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right)\right] \int_R \exp\left(-\frac{itu}{b_1}\right) \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) dt \\
 &\stackrel{\xi=\frac{t-b}{a}}{=} \frac{\sqrt{a}}{\sqrt{i2\pi b_1}} \exp\left[\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) - \frac{ibu}{b_1}\right] \int_R \exp\left(-\frac{iau\xi}{b_1}\right) \psi(\xi) d\xi \\
 &= \frac{\sqrt{a}}{\sqrt{i2\pi b_1}} \exp\left[\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) - \frac{ibu}{b_1} - \frac{i}{2}\frac{d_1}{b_1}a^2u^2\right] \\
 &\quad \times \int_R \exp\left(i \frac{a_1 \xi^2 + d_1(au)^2 - 2\xi \cdot (au)}{2b_1}\right) \left[\psi(\xi) \exp\left(-\frac{i}{2}\frac{a_1}{b_1}\xi^2\right)\right] d\xi \\
 &= \sqrt{a} \exp\left[\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) - \frac{ibu}{b_1} - \frac{i}{2}\frac{d_1}{b_1}a^2u^2\right] \\
 &\quad \times \mathcal{L}^{A_1}\left[\exp\left(-\frac{i}{2}\frac{a_1}{b_1}\xi^2\right)\psi(\xi)\right](au).
 \end{aligned}$$

Hence, the LCWT can also be written as

$$\begin{aligned}
 W_f^{A_1}(a, b) &= \int_R f(t) \overline{\psi_{a,b,A_1}(t)} dt \\
 &= \int_R \mathcal{L}^{A_1}[f(t)](u) \overline{\mathcal{L}^{A_1}[\psi_{a,b,A_1}(t)](u)} du \\
 &= \sqrt{a} \int_R \mathcal{L}^{A_1}[f(t)](u) \exp\left[-\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) + \frac{ibu}{b_1} + \frac{i}{2}\frac{d_1}{b_1}a^2u^2\right] \\
 &\quad \times \overline{\mathcal{L}^{A_1}\left[\exp\left(-\frac{i}{2}\frac{a_1}{b_1}\xi^2\right)\psi(\xi)\right](au)} du \\
 &= \sqrt{a} \int_R L_f^{A_1}(u) \exp\left[-\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) + \frac{ibu}{b_1} + \frac{i}{2}\frac{d_1}{b_1}a^2u^2\right] \\
 &\quad \times \overline{\mathcal{L}^{A_1}\left[\exp\left(-\frac{i}{2}\frac{a_1}{b_1}\xi^2\right)\psi(\xi)\right](au)} du.
 \end{aligned}$$

In follows, we will show that the above definition is equivalent to Eq. (3.4).

$$\begin{aligned}
 & \sqrt{a} \int_R \mathcal{L}^{A_1}[f(t)](u) \exp\left[\frac{i}{2}\left(\frac{d_1}{b_1}u^2 + \frac{a_1}{b_1}b^2\right) + \frac{ibu}{b_1} + \frac{i}{2}\frac{d_1}{b_1}a^2u^2\right] \\
 &\quad \times \overline{\mathcal{L}^{A_1}\left[\exp\left(-\frac{i}{2}\frac{a_2}{b_2}\xi^2\right)\psi(\xi)\right](au)} du
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\sqrt{a}}{\sqrt{-i2\pi b_1}} \int_R \mathcal{L}^{A_1}[f(t)](u) \exp \left[-\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) + \frac{ibu}{b_1} \right] \\
&\quad \times \overline{\int_R \exp \left(-\frac{iau\xi}{b_1} \right) \psi(\xi) d\xi du} \\
&= \sqrt{a} \int_R \mathcal{L}^{A_1}[f(t)](u) K_{A_1^{-1}}(b, u) \overline{\int_R \exp \left(-\frac{iau\xi}{b_1} \right) \psi(\xi) d\xi du} \\
&= \sqrt{2\pi a} \int_R \mathcal{L}^{A_1}[f(t)](u) \overline{F_\psi \left(\frac{au}{b_1} \right)} K_{A_1^{-1}}(b, u) du.
\end{aligned}$$

□

3.2. Some basic properties of LCWT

In this section, some basic properties of LCWT are derived.

Theorem 3.2. Assume $\forall f(t) \in \mathcal{S}_{A_1}(R)$ and $\Delta_b = \frac{d}{db} - ib\frac{a_1}{b_1}$, then

- (i) $\Delta_t^\beta \overline{\psi_{a,b,A_1}(t)} = D^\beta \overline{\psi_{a,b}(t)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}}$.
- (ii) $(\Delta_b^*)^\beta \overline{\psi_{a,b,A_1}(t)} = D^\beta \overline{\psi_{a,b}(t)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}}$.
- (iii) $(\Delta_b^*)^\beta \overline{\psi_{a,b,A_1}(t)} = \Delta_t^\beta \overline{\psi_{a,b,A_1}(t)}$.
- (iv) $\int_R \overline{\psi_{a,b,A_1}(t)} [(\Delta_t^*)^\beta f(t)] dt = \int_R \Delta_t^\beta \overline{\psi_{a,b,A_1}(t)} f(t) dt$.
- (v) $\mathcal{W}^{A_1}[(\Delta_b^*)^\beta f(t)](a, b) = (\Delta_b^*)^\beta \mathcal{W}^{A_1}[f(t)](a, b)$.
- (vi) $\mathcal{W}^{A_1}[t^\alpha \Delta_t^* f(t)](a, b) = \Delta_b^* \mathcal{W}^{A_1}[t^\alpha f(t)](a, b) + \mathcal{W}^{A_1}[\alpha t^{\alpha-1} f(t)](a, b)$.
- (vii) $\int_{R^+} \int_R W_f^{A_1}(a, b) \overline{[W_g^{A_1}(a, b)]} \frac{1}{a^2} da db = 2\pi b_1 C_{\Psi, A_1} \int_{-\infty}^{+\infty} f(t) \overline{g(t)} dt$.
- (viii) $\int_R \int_{R^+} |W_f^{A_1}(a, b)|^2 \frac{1}{a^2} da db = 2\pi b_1 C_{\Psi, A_1} \int_R |f(t)|^2 dt$.

Proof. (i) According to the definition of $\psi_{a,b,A_1}(t)$, then

$$\begin{aligned}
\Delta_t \overline{\psi_{a,b,A_1}(t)} &= \left(\frac{d}{dt} - it \frac{a_1}{b_1} \right) \overline{\psi_{a,b,A_1}(t)} \\
&= \left(\frac{d}{dt} - it \frac{a_1}{b_1} \right) \frac{1}{\sqrt{a}} \overline{\psi \left(\frac{t-b}{a} \right)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}} \\
&= \frac{1}{a\sqrt{a}} \overline{\psi' \left(\frac{t-b}{a} \right)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}} + \frac{1}{\sqrt{a}} \overline{\psi \left(\frac{t-b}{a} \right)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}} \\
&\quad \cdot \left(it \frac{a_1}{b_1} - it \frac{a_1}{b_1} \right) \\
&= \frac{1}{a\sqrt{a}} \overline{\psi' \left(\frac{t-b}{a} \right)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}} \\
&= D \overline{\psi_{a,b}(t)} e^{\frac{i}{2}(t^2-b^2) \cdot \frac{a_1}{b_1}}.
\end{aligned}$$

The final result can be obtained by continuing in this way.

- (ii) The proof is similar to that of (i).
- (iii) It can be derived easily based on the results of (i) and (ii).
- (iv) By using the section integral, we have

$$\begin{aligned}
 & \int_R \overline{\psi_{a,b,A_1}(t)} [\Delta_t^* f(t)] dt \\
 &= - \int_R \overline{\psi_{a,b,A_1}(t)} \left(\frac{d}{dt} + it \frac{a_1}{b_1} \right) f(t) dt \\
 &= - \int_R \overline{\psi_{a,b,A_1}(t)} df(t) - \int_R \left(it \frac{a_1}{b_1} \right) \overline{\psi_{a,b,A_1}(t)} f(t) dt \\
 &= \int_R \overline{\psi'_{a,b,A_1}(t)} f(t) dt - \int_R \left(it \frac{a_1}{b_1} \right) \overline{\psi_{a,b,A_1}(t)} f(t) dt \\
 &= \int_R \Delta_t \overline{\psi_{a,b,A_1}(t)} f(t) dt.
 \end{aligned}$$

Continuing in this way, then

$$\int_R \overline{\psi_{a,b,A_1}(t)} [(\Delta_t^*)^\beta f(t)] dt = \int_R \Delta_t^\beta \overline{\psi_{a,b,A_1}(t)} f(t) dt. \quad (3.6)$$

- (v) Based on the results of (iii) and (iv), then

$$\begin{aligned}
 \mathcal{W}^{A_1}[(\Delta_t^*)^\beta f(t)](a, b) &= \int_R \overline{\psi_{a,b,A_1}(t)} [(\Delta_t^*)^\beta f(t)] dt \\
 &= \int_R \Delta_t^\beta \overline{\psi_{a,b,A_1}(t)} f(t) dt \\
 &= \int_R (\Delta_b^*)^\beta \overline{\psi_{a,b,A_1}(t)} f(t) dt \\
 &= (\Delta_b^*)^\beta \int_R \overline{\psi_{a,b,A_1}(t)} f(t) dt \\
 &= (\Delta_b^*)^\beta \mathcal{W}^{A_1}[f(t)](a, b).
 \end{aligned}$$

- (vi) Based on the above results, we have

$$\begin{aligned}
 & \mathcal{W}^{A_1}[t^\alpha \Delta_t^* f(t)](a, b) \\
 &= \int_R \overline{\psi_{a,b,A_1}(t)} [t^\alpha \Delta_t^* f(t)] dt \\
 &= \int_R \overline{\psi_{a,b,A_1}(t)} [\Delta_t^*(t^\alpha f(t)) + \alpha t^{\alpha-1} f(t)] dt \\
 &= \int_R \Delta_t \overline{\psi_{a,b,A_1}(t)} (t^\alpha f(t)) dt + \int_R \overline{\psi_{a,b,A_1}(t)} [\alpha t^{\alpha-1} f(t)] dt \\
 &= \int_R \Delta_b^* \overline{\psi_{a,b,A_1}(t)} (t^\alpha f(t)) dt + \int_R \overline{\psi_{a,b,A_1}(t)} [\alpha t^{\alpha-1} f(t)] dt \\
 &= \Delta_b^* \mathcal{W}^{A_1}[t^\alpha f(t)](a, b) + \mathcal{W}^{A_1}[\alpha t^{\alpha-1} f(t)](a, b).
 \end{aligned}$$

(vii) Since

$$W_f^{A_1}(a, b) = \sqrt{a} \int_R L_f^{A_1}(u) \exp \left[-\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) + \frac{ibu}{b_1} + \frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right] \\ \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au)} du$$

$$W_g^{A_1}(a, b) = \sqrt{a} \int_R L_g^{A_1}(v) \exp \left[-\frac{i}{2} \left(\frac{d_1}{b_1} v^2 + \frac{a_1}{b_1} b^2 \right) + \frac{ibv}{b_1} + \frac{i}{2} \frac{d_1}{b_1} a^2 v^2 \right] \\ \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (av)} dv$$

then,

$$\begin{aligned} & \int_{R^+} \int_R W_f^{A_1}(a, b) \overline{W_g^{A_1}(a, b)} \frac{1}{a^2} da db \\ &= \int_{R^+} \int_R \left[\int_R L_f^{A_1}(u) \exp \left[-\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) + \frac{ibu}{b_1} + \frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right] \right. \\ & \quad \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au)} du \Big] \\ & \quad \cdot \left[\int_R \overline{L_g^{A_1}(v)} \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} v^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibv}{b_1} - \frac{i}{2} \frac{d_1}{b_1} a^2 v^2 \right] \right. \\ & \quad \times \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (av) dv \Big] \frac{1}{a} da db \\ &= \int_{R^+} \int_R \int_R L_f^{A_1}(u) \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} v^2 - \frac{a_1}{b_1} u^2 \right) - \frac{i}{2} \frac{d_1}{b_1} a^2 (v^2 - u^2) \right] \\ & \quad \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au)} \cdot \overline{L_g^{A_1}(v)} \mathcal{L}^{A_1} \\ & \quad \times \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (av) \int_R \exp \left(\frac{ib(u-v)}{b_1} \right) db \frac{1}{a} du dv da \\ &= 2\pi b_1 \int_{R^+} \int_R \int_R L_f^{A_1}(u) \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} v^2 - \frac{a_1}{b_1} u^2 \right) - \frac{i}{2} \frac{d_1}{b_1} a^2 (v^2 - u^2) \right] \\ & \quad \times \overline{\mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au)} \\ & \quad \cdot \overline{L_g^{A_1}(v)} \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (av) \delta(u-v) \frac{1}{a} du dv da \end{aligned}$$

$$\begin{aligned}
 &= 2\pi b_1 \int_R L_f^{A_1}(u) \overline{L_g^{A_1}(u)} \\
 &\quad \times \left[\int_{R^+} \left| \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) \right|^2 \cdot \frac{1}{a} da \right] du \\
 &= 2\pi b_1 C_{\Psi, A_1} \int_R L_f^{A_1}(u) \overline{L_g^{A_1}(u)} dt \\
 &= 2\pi b_1 C_{\Psi, A_1} \int_R f(t) \overline{g(t)} dt.
 \end{aligned}$$

(viii) Let $f(t) = g(t)$ in (vii), then the result can be obtained. $\square$

3.3. LCWT on some test function spaces

The continuous and boundness of WT and FRWT have been studied on some test function spaces, such as the Gelfand–Shilov spaces^{10,11,18,19} and generalized Sobolev spaces.^{14,16,17} These studies enlarge the application fields of WT and FRWT.

3.3.1. LCWT on function space L^{p, A_1}

Definition 3.2. For $1 \leq p < \infty$, the generalized test function space L^{p, A_1} is defined as

$$L^{p, A_1} = \{\phi \in C^\infty \mid t^\alpha (\Delta_t^*)^\beta \phi(t) \in L^p, \forall \alpha, \beta \in N\}. \quad (3.7)$$

For all $\phi \in L^{p, A_1}$, its norm is defined as

$$\|\phi\|_{L^{p, A_1}} = \|t^\alpha (\Delta_t^*)^\beta \phi(t)\|_{L^p} (\forall \alpha, \beta \in N). \quad (3.8)$$

Definition 3.3. The generalized test function space G^{p, A_1} is defined as

$$\begin{aligned}
 G^{p, A_1} = \left\{ \varphi(a, b) \in C^\infty(R^+ \times R) \mid \right. \\
 \left. \left(\int_R \left| \int_{R^+} a^\sigma b^\alpha D_a^\gamma (\Delta_b^*)^\beta [\sqrt{a} \varphi(a, b)] da \right|^p db \right)^{\frac{1}{p}} < \infty \right\}, \quad (3.9)
 \end{aligned}$$

where $\sigma, \alpha, \beta, \gamma \in N$ satisfying $\beta + \gamma - \sigma - 2 \geq 0$. For all $\varphi(a, b) \in G^{p, A_1}$, its norm is defined as

$$\|\varphi(a, b)\|_{G^{p, A_1}} = \left(\int_R \left| \int_{R^+} a^\sigma b^\alpha D_a^\gamma (\Delta_b^*)^\beta [\sqrt{a} \varphi(a, b)] da \right|^p db \right)^{\frac{1}{p}}. \quad (3.10)$$

Theorem 3.3. Assume $\phi \in L^{p, A_1}$ and $\psi \in \mathcal{S}_{A_1}(R)$ satisfy the condition

$$C'_{\psi, A_1} = \left| \left(\int_{R^+} \tau^\sigma D_\tau^\gamma \left[\tau \exp\left(\frac{i}{2} \frac{d_1}{b_1} \tau^2\right) \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (\tau) \right] d\tau \right) \right| < \infty,$$

then, the LCWT operator $\mathcal{W}^{A_1} : L^{p, A_1} \rightarrow G^{p, A_1}$ is continuous.

Proof.

$$\begin{aligned}
I &= a^\sigma b^\alpha D_a^\gamma (\Delta_b^*)^\beta \overline{\sqrt{a} W_\phi^{A_1}(a, b)} \\
&= a^\sigma b^\alpha D_a^\gamma (\Delta_b^*)^\beta \int_R a L_\phi^{A_1}(u) \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibu}{b_1} - \frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right] \\
&\quad \times \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au) \, du \\
&= a^\sigma b^\alpha D_a^\gamma \int_R a L_\phi^{A_1}(u) \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibu}{b_1} - \frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right] \left(-\frac{iu}{b_1} \right)^\beta \\
&\quad \times \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au) \, du.
\end{aligned}$$

Hence,

$$\begin{aligned}
\int_{R^+} I \, da &= \left(-\frac{i}{b_1} \right)^\beta b^\alpha \int_R \left(\int_{R^+} a^\sigma D_a^\gamma \left[a \exp \left(\frac{i}{2} \frac{d_1}{b_1} a^2 u^2 \right) \right. \right. \\
&\quad \times \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (au) \left. \left. \right] \, da \right) \cdot u^\beta L_\phi^{A_1}(u) \\
&\quad \times \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibu}{b_1} \right] \, du \\
&\stackrel{\tau=au}{=} \left(-\frac{i}{b_1} \right)^\beta b^\alpha \int_R \left(\int_{R^+} \tau^\sigma D_\tau^\gamma \left[\tau \exp \left(\frac{i}{2} \frac{d_1}{b_1} \tau^2 \right) \right. \right. \\
&\quad \times \mathcal{L}^{A_1} \left[\exp \left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2 \right) \psi(\xi) \right] (\tau) \left. \left. \right] \, d\tau \right) \cdot u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u) \\
&\quad \times \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibu}{b_1} \right] \, du.
\end{aligned}$$

Moreover, we have

$$\begin{aligned}
\left| \int_{R^+} I \, da \right| &\leq C'_{\psi, A_1} \left| \frac{1}{b_1} \right|^\beta \\
&\quad \times \left| b^\alpha \int_R u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u) \exp \left[\frac{i}{2} \left(\frac{d_1}{b_1} u^2 + \frac{a_1}{b_1} b^2 \right) - \frac{ibu}{b_1} \right] \, du \right| \\
&\leq C'_{\psi, A_1} \left| \frac{1}{b_1} \right|^\beta \cdot |b^\alpha \sqrt{i 2\pi b_1} \mathcal{L}[u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u)](b)| \\
&\leq \sqrt{2\pi} C'_{\psi, A_1} \left| \frac{1}{b_1} \right|^{\beta-\alpha-1/2} |\mathcal{L}[(\Delta_u^*)^\alpha (u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u))](b)|
\end{aligned}$$

then, based on the Riesz–Thorin interpolation formula,⁶ we have

$$\begin{aligned}
 & \left(\int_R \left| \int_{R^+} I \, da \right|^p db \right)^{\frac{1}{p}} \\
 & \leq \sqrt{2\pi} C'_{\psi, A_1} \left| \frac{1}{b_1} \right|^{\beta-\alpha-1/2} \left(\int_R |\mathcal{L}[(\Delta_u^*)^\alpha (u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u))](b)|^p db \right) \\
 & \leq C''_{\psi, A_1} \left(\int_R |[(\Delta_u^*)^\alpha (u^{\beta+\gamma-\sigma-2} L_\phi^{A_1}(u))](u)|^q du \right)^{\frac{1}{q}} \\
 & = C''_{\psi, A_1} b_1^{\beta+\gamma-\sigma-2} \left(\int_R |[(\Delta_u^*)^\alpha \mathcal{L}((\Delta_t^*)^{\beta+\gamma-\sigma-2} \phi(t))](u)|^q du \right)^{\frac{1}{q}} \\
 & = C''_{\psi, A_1} b_1^{\beta+\gamma-\alpha-\sigma-2} \left(\int_R |\overline{\mathcal{L}[t^\alpha (\Delta_t^*)^{\beta+\gamma-\sigma-2} \phi(t)](u)}|^q du \right)^{\frac{1}{q}} \\
 & = C''_{\psi, A_1} b_1^{\beta+\gamma-\alpha-\sigma-2} \left(\int_R |\mathcal{L}[t^\alpha (\Delta_t^*)^{\beta+\gamma-\sigma-2} \phi(t)](u)|^q du \right)^{\frac{1}{q}} \\
 & \leq C'''_{\psi, A_1} |b_1^{\beta+\gamma-\alpha-\sigma-2}| \left(\int_R |t^\alpha (\Delta_t^*)^{\beta+\gamma-\sigma-2} \phi(t)|^p dt \right)^{\frac{1}{p}} \\
 & = C'''_{\psi, A_1} |b_1^{\beta+\gamma-\alpha-\sigma-2}| \|\phi\|_{L^{p, A_1}}. \quad \square
 \end{aligned}$$

3.3.2. LCWT on generalized Sobolev space $H_{A_1}^{s,p}$

Definition 3.4. For $1 \leq p < \infty$ and $-\infty \leq s < \infty$, the generalized Sobolev space $H_{A_1}^{s,p}$ is defined as

$$H_{A_1}^{s,p} = \{\phi(t) \in \mathcal{S}'_{A_1}(R) \mid u^{\frac{s}{p}} L_\phi^{A_1}(u) \in L^p(R)\}.$$

Its norm is defined as $\|\phi\|_{H_{A_1}^{s,p}} = \|u^{\frac{s}{p}} L_\phi^{A_1}(u)\|_{L^p}$.

Definition 3.5. The generalized Sobolev space $H_{q, A_1}^{s,p}$ includes all measurable function $\varphi(a, b)$ satisfying

$$\|\varphi(a, b)\|_{H_{q, A_1}^{s,p}} = \left[\int_{R^+} \left(\int_R |\varphi(a, b)|^p db \right)^{q/p} a^{-s-1} da \right]^{1/q} < \infty.$$

Theorem 3.4. Assume that the analyzing wavelet satisfies the following condition:

$$C_{\psi, A_1}^{s,q} = \int_{R^+} \frac{\left| \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) \right|^q}{a|u|^s} \frac{da}{a} < \infty$$

then, the LCWT is a bounded linear operator from $H_{A_1}^{s,p}(R)$ into $H_{q, A_1}^{s,p}(R \times R^+)$ for $1 \leq p \leq 2, q = \frac{p}{p-1}, s \in R$.

Proof. Because

$$\mathcal{L}^{A_1}[W_f^{A_1}(a, b)] = \sqrt{2\pi a} \exp\left(\frac{i}{2} \frac{a_1}{b_1} a^2 u^2\right) \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u). \quad (3.11)$$

Using the Riesz–Thorin interpolation formula for LCT, we have

$$\begin{aligned} & \left(\int_R \left| \sqrt{2\pi a} \exp\left(\frac{i}{2} \frac{a_1}{b_1} a^2 u^2\right) \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right|^q du \right)^{1/q} \\ &= \left(\int_R \left| \mathcal{L}^{A_1}[W_f^{A_1}(a, b)] \right|^q du \right)^{1/q} \\ &\leq C_{p, A_1} \left(\int_R \left| W_f^{A_1}(a, b) \right|^p du \right)^{1/p}, \end{aligned}$$

where $C_{p, A_1} > 0$ is a constant. Hence,

$$\begin{aligned} & \int_{R^+} \left(\int_R \left| \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right|^q du \right) a^{-s-1} da \\ &= C_{\psi, A_1}^{s, q} \int_R |u|^s |L_f^{A_1}(u)|^q du \\ &\leq \left(\frac{C_{p, A_1}}{\sqrt{2\pi a}} \right)^q \int_{R^+} \left(\int_R \left| W_f^{A_1}(a, b) \right|^p db \right)^{q/p} a^{-s-1} da \\ &\leq \left(\frac{C_{p, A_1}}{\sqrt{2\pi a}} \right)^q \|W_f^{A_1}(a, b)\|_{H_{q, A_1}^{s, p}}^q. \end{aligned}$$

Moreover, we have

$$\|f\|_{H_{A_1}^{s, q}} \leq \frac{C_{p, A_1}}{\sqrt{2\pi a} (C_{\psi, A_1}^{s, q})^{1/q}} \|W_f^{A_1}(a, b)\|_{H_{q, A_1}^{s, p}}. \quad (3.12)$$

Furthermore, based on the Riesz–Thorin interpolation formula⁶ and Eq. (3.11), then

$$\begin{aligned} & \left(\int_R \left| W_f^{A_1}(a, b) \right|^q db \right)^{1/q} \\ &= \left(\int_R \left| \frac{\sqrt{a}}{\sqrt{i2\pi b_1}} \mathcal{L}^{A_1^{-1}} \left[\exp\left(\frac{i}{2} \frac{a_1}{b_1} a^2 u^2\right) \right. \right. \right. \\ &\quad \times \left. \left. \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right] (b) \right|^q db \right)^{1/q} \\ &\leq \frac{C'_{p, A_1}}{\sqrt{i2\pi b_1}} \left(\int_R \left| \exp\left(\frac{i}{2} \frac{a_1}{b_1} a^2 u^2\right) \right. \right. \\ &\quad \times \left. \left. \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right|^p du \right)^{1/p}, \end{aligned}$$

where $C'_{p,A_1} > 0$ is a constant. Then,

$$\begin{aligned} & \int_{R^+} \left(\int_R |W_f^{A_1}(a, b)|^q db \right)^{p/q} a^{-s-1} da \\ & \leq \left(\frac{C'_{p,A_1}}{\sqrt{i2\pi b_1}} \right)^p \left(\int_R \left| \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right|^p du \right) da \\ & \leq C_{\psi,A_1}^{s,q} \left(\frac{C'_{p,A_1}}{\sqrt{i2\pi b_1}} \right)^p \left(\int_R \left| \mathcal{L}^{A_1} \left[\exp\left(-\frac{i}{2} \frac{a_1}{b_1} \xi^2\right) \psi(\xi) \right] (au) L_f^{A_1}(u) \right|^p du \right) da \\ & \leq C_{\psi,A_1}^{s,q} \left(\frac{C'_{p,A_1}}{\sqrt{i2\pi b_1}} \right)^p \int_R |u|^s |L_f^{A_1}(u)|^q du. \end{aligned}$$

Therefore, we obtain

$$\|W_f^{A_1}(a, b)\|_{H_{q,A_1}^{s,p}} \leq (C_{\psi,A_1}^{s,q})^{1/p} \left(\frac{C'_{p,A_1}}{\sqrt{i2\pi b_1}} \right) \|f\|_{H_{A_1}^{s,q}}. \quad (3.13)$$

Based on the Eqs. (3.12) and (3.13), it can be concluded that LCWT is a bounded linear operator from $H_{A_1}^{s,p}(R)$ into $H_{q,A_1}^{s,p}(R \times R^+)$. $\square$

4. Conclusions

In this paper, the definition of FT on $\mathcal{S}(R)$ is introduced first, as well as some properties. Moreover, some properties of LCT on $\mathcal{S}_{A_1}(R)$ are studied. Based on the convolution theorem in LCT domain, we propose an equivalent definition of LCWT associated with LCT and further study some properties of LCWT on $\mathcal{S}_{A_1}(R)$. According to these properties, the continuity of LCWT on some test function space is detected.

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References

1. M. J. Bastiaans and T. Alieva, *The Linear Canonical Transformation* (Springer, New York, 2016).
2. C. Bouzar and T. Saidi, Fourier analysis of generalized functions, *Integral Transforms Spec. Funct.* **22**(4–5) (2011) 337–344.
3. Q. Feng and B. Z. Li, Convolution and correlation theorems for the two-dimensional linear canonical transform and its applications, *IET Signal Process.* **10**(2) (2016) 125–132.
4. N. Goel *et al.*, Multiplicative filtering in the linear canonical transform domain, *IET Signal Process.* **10**(2) (2016) 173–181.

5. N. Goel and K. Singh, Modified correlation theorem for the linear canonical transform with representation transformation in quantum mechanics, *Signal Image Video Process.* **8**(3) (2014) 595–601.
6. L. Grafakos, *Classical Fourier Analysis* (Springer, New York, 2008).
7. V. Karunakaran and C. Ganesan, Fourier transform on integrable Boehmians, *Integral Transforms Spec. Funct.* **20**(12) (2009) 937–941.
8. A. A. Kilbas *et al.*, Fractional Fourier transform in the framework of fractional calculus operators, *Integral Transforms Spec. Funct.* **21**(10) (2010) 779–795.
9. R. S. Pathak, A. Prasad and M. Kumar, Fractional Fourier transform of tempered distributions and generalized pseudo-differential operator, *J. Pseudo.-Differ. Oper.* **3**(2) (2012) 239–254.
10. R. S. Pathak and S. K. Singh, The wavelet transform on spaces of type S, *Proc. Roy. Soc. Enidburgh. A* **136**(4) (2006) 837–850.
11. S. Pilipovi, D. Raki, N. Teofanov and J. Vindas, The wavelet transforms in Gelfand–Shilov spaces, *Collect. Math.* **67**(3) (2016) 443–460.
12. A. Prasad *et al.*, The generalized continuous wavelet transform associated with the fractional Fourier transform, *J. Comput. Appl. Math.* **259** (2014) 660–671.
13. A. Prasad and M. Kumar, Product of two generalized pseudo-differential operators involving fractional Fourier transform, *J. Pseudo.-Differ. Oper.* **2**(3) (2011) 355–365.
14. A. Prasad and P. Kumar, The continuous fractional wavelet transform on generalized weighted Sobolev spaces, *Asian-European J. Math.* **8**(3) (2015) 1550054.
15. A. Prasad and P. Kumar, Composition of the continuous fractional wavelet transforms, *Natl. Acad. Sci. Lett.* **39**(2) (2016) 115–120.
16. A. Prasad and P. Kumar, The continuous fractional wavelet transform on a generalized Sobolev space, *Int. J. Wavelets Multiresolut. Inf. Process.* **14**(6) (2016) 1650046.
17. A. Prasad and P. Kumar, Fractional continuous wavelet transform on some function spaces, *Proc. Natl. Acad. Sci. India A* **86**(1) (2016) 57–64.
18. A. Prasad and A. Mahato, The fractional wavelet transform on spaces of type S, *Integral Transforms Spec. Funct.* **23**(4) (2012) 237–249.
19. A. Prasad and A. Mahato, The fractional wavelet transform on spaces of type W, *Integral Transforms Spec. Funct.* **24**(3) (2013) 239–250.
20. J. Shi *et al.*, Extrapolation of bandlimited signals in linear canonical transform domain, *IEEE Trans. Signal Process.* **60**(3) (2012) 1502–1508.
21. J. Shi, N. T. Zhang and X. P. Liu, A novel fractional wavelet transform and its applications, *Sci. China Inform. Sci.* **55**(6) (2012) 1270–1279.
22. D. Wei and Y. Li, Generalized wavelet transform based on the convolution operator in the linear canonical transform domain, *Optik* **125**(16) (2014) 4491–4496.
23. T. Z. Xu and B. Z. Li, *The Linear Canonical Transform and Its Applications* (Science Press, Beijing, 2013).